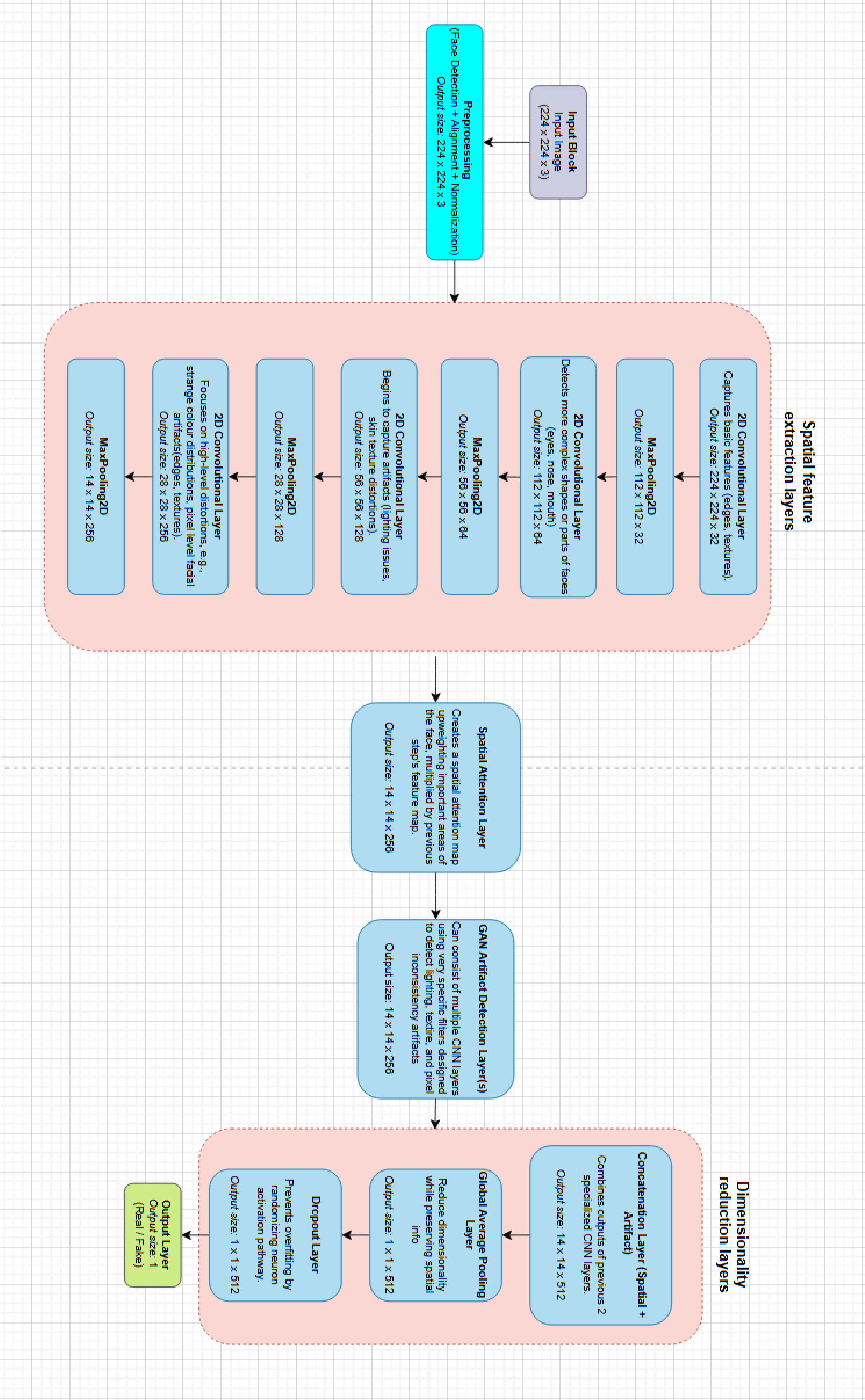
CNN architecture diagram



1. Input Layer
    a. Shape Handling
        i. Height x width x colour channels
    b. Pre-processing
        i. Face detection: using a pre-trained model to detect the face and crop the image if the face only occupies part of the image
        ii. Face alignment: using dlib or DeepFace to ensure facial landmarks (eyes, mouth, nose, etc.) are consistently positioned
        iii. Normalization: ensure pixel values are between 0 and 1
2. Spatial feature extraction layers (CNN)
    a. Simple Conv2D layers for edge / corner / texture detection to begin, but eventually focus on artifacts and inconsistencies in texture, lighting, and facial features.
        i. Conv2D(32): Captures basic features (edges, textures).
        ii. Conv2D(64): Detects more complex shapes or parts of faces (eyes, nose, mouth).
        iii. Conv2D(128): Begins to capture artifacts (lighting issues, skin texture distortions).
        iv. Conv2D(256): Focuses on high-level distortions specific to deepfake images, such as unnatural lighting, strange color distributions, or pixel-level artifacts in the face.
3. Facial region attention layer
    a. Deepfake images often have anomalies in very specific regions, e.g., mouth, eyes, jawline. **Attention mechanisms** bring focus to these areas, getting the model to pay more attention to the regions most prone to being manipulated.
        i. Do this through an attention map that applies more weight to findings/distortions in key regions, i.e., the face, more specifically, eyes, mouth, and facial symmetry.
4. Deepfake-specific artifact detection layer
    a. Layer that hones in on deepfake-specific visual artifacts:
        i. Irregularities in skin texture: e.g., skin might be overly smooth or have inconsistent pores
        ii. Unnatural lighting / shadows: generative adversarial network (GAN)-generated images can have lighting inconsistencies, e.g., improper angle of shadows with respect to object casting the shadow.
        iii. Expression misaligned with facial movements
5. Concatenation layer to merge the output of the facial region attention layer with the artifact-detection layer
6. Global feature extraction layer
    a. Global average pooling to reduce dimensionality but preserve spatial info
7. Fully connected layer with 50% dropout (to avoid overfitting)
8. Final output layer

Explanation of Size Changes:

1. **Input Image (224x224x3)**: This is the original image with 224 pixels in height, 224 pixels in width, and 3 color channels (RGB).

2. **Conv2D (64 filters, 3x3)**: After applying the first convolution, the output size is still **224x224** because padding is used to preserve the spatial dimensions. The number of channels increases to **64** due to the 64 filters.

3. **MaxPooling2D (2x2)**: The pooling layer reduces the spatial dimensions by a factor of 2, so the height and width of the feature map become **112x112** while the number of channels (filters) remains **64**.

4. **Conv2D (128 filters, 3x3)**: The second convolution with 128 filters gives an output of size **112x112** (since no padding is used), and the number of channels increases to **128**.

5. **MaxPooling2D (2x2)**: Pooling again reduces the spatial dimensions to **56x56** while keeping the 128 channels.

6. **Conv2D (256 filters, 3x3)**: Another convolution layer increases the channel count to **256** while the spatial dimensions stay at **56x56**.

7. **MaxPooling2D (2x2)**: After this pooling, the spatial dimensions reduce to **28x28**, with the number of channels remaining **256**.

8. **Conv2D (512 filters, 3x3)**: This convolution layer increases the number of filters to **512**, while the spatial dimensions remain **28x28**.

9. **MaxPooling2D (2x2)**: Pooling once more reduces the spatial dimensions to **14x14**, while the number of channels remains **512**.

10. **Spatial Attention Layer**: This layer is used to focus on important facial regions (like eyes, nose, etc.), but it doesn't change the size of the feature map. Therefore, the output remains **14x14x512**.

11. **Artifact Detection Layer (Conv2D, 512 filters)**: This layer detects artifacts specific to deepfakes. It doesn't change the size, so the output remains **14x14x512**.

12. **Concatenate Layers**: After concatenating the output of the spatial attention and artifact detection layers, the number of channels doubles to **1024**, while the spatial dimensions remain **14x14**.

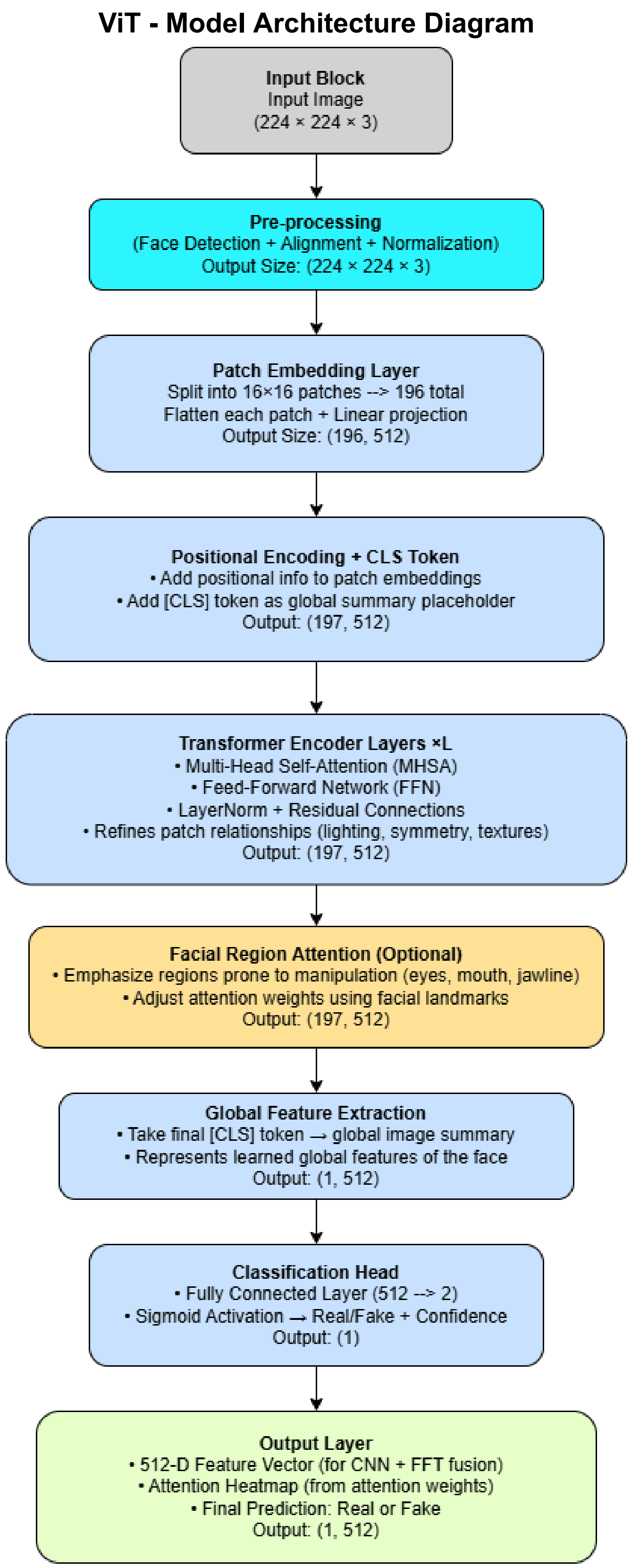
13. **Global Average Pooling**: This layer reduces the spatial dimensions to **1x1** by averaging the values across the entire feature map. The output has size **1x1x1024**,

representing a condensed feature vector.

14. **Dense Layer (1024 units)**: The dense layer doesn't change the size of the feature map, but it reduces it to a single feature vector of size **1x1x1024**.

15. **Dropout (0.5)**: The **Dropout** layer randomly disables 50% of the neurons, but doesn't change the size of the layer. It only affects the training process by preventing overfitting.

16. **Output Layer (Sigmoid Activation)**: The final output is a single value representing the prediction: **1 (real)** or **0 (fake)**, which is the output of the sigmoid activation function. The size is **1**.


ViT - Model Architecture Diagram
Input Block
Input Image
(224 × 224 × 3)
Pre-processing
(Face Detection + Alignment + Normalization)
Output Size: (224 × 224 × 3)
Patch Embedding Layer
Split into 16×16 patches --> 196 total
Flatten each patch + Linear projection
Output Size: (196, 512)
Positional Encoding + CLS Token
• Add positional info to patch embeddings
• Add [CLS] token as global summary placeholder
Output: (197, 512)
Transformer Encoder Layers ×L
• Multi-Head Self-Attention (MHSA)
• Feed-Forward Network (FFN)
• LayerNorm + Residual Connections
• Refines patch relationships (lighting, symmetry, textures)
Output: (197, 512)
Facial Region Attention (Optional)
• Emphasize regions prone to manipulation (eyes, mouth, jawline)
• Adjust attention weights using facial landmarks
Output: (197, 512)
Global Feature Extraction
• Take final [CLS] token → global image summary
• Represents learned global features of the face
Output: (1, 512)
Classification Head
• Fully Connected Layer (512 --> 2)
• Sigmoid Activation → Real/Fake + Confidence
Output: (1)
Output Layer
• 512-D Feature Vector (for CNN + FFT fusion)
• Attention Heatmap (from attention weights)
• Final Prediction: Real or Fake
Output: (1, 512)

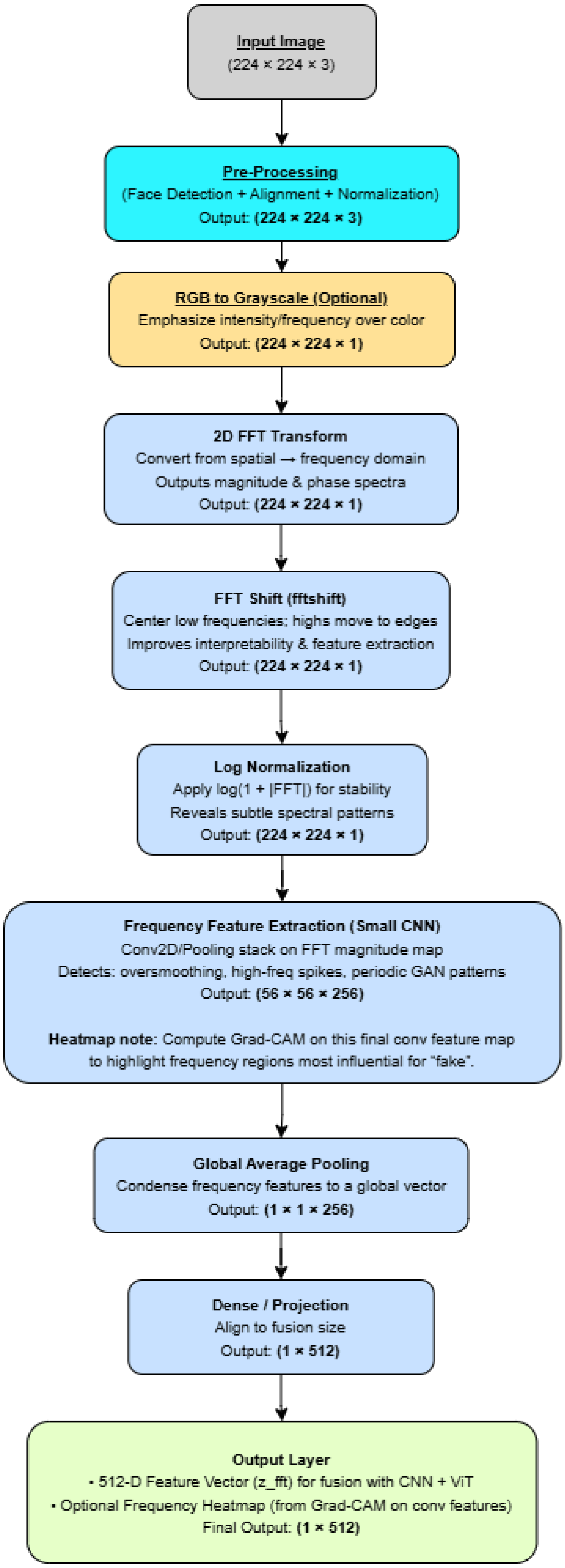
ViT - Diagram Explanation

1. Input Layer
    - Shape Handling
        - Input shape: **Height × Width × Color Channels (224 × 224 × 3)**
        - Accepts the same preprocessed face image used by CNN and FFT branches.
        - Image has already gone through shared preprocessing: face detection, alignment, resize, and normalization.
2. Patch Embedding
    - Aligned image is divided into small square patches (16×16 pixels each).
    - Each patch is flattened into a **1D vector**, passed through a linear layer that converts it into an **embedding** (a list of numbers describing that patch) - this is the vector the ViT uses.
    - You now have around 196 patches (if 224×224 image = 14 16×16 patches).
3. Positional Encoding
    - Transformers don't know where each patch came from (top-left, bottom-right, etc.).
    - Positional encoding added to each patch embedding, so the model knows where that patch is on the face.
    - CLS token (classification token) is also added at beginning of sequence, acting as a special "summary" placeholder for the entire image (**puts the pieces together**).
4. Transformer Encoder Layers
    - Main body of the ViT is made up of repeated layers (ex: 6–12 layers) that contain:
        - Multi-Head Self-Attention (MHSA): Each patch compares itself with all other patches to understand relationships (ex: is lighting on the left cheek consistent with the right cheek?).
        - Feed-Forward Network (FFN): Mini NN that refines the information in each patch (compressing newly learned info).
        - Layer Normalization + Residual Connections: Help stabilize and speed up training.
        - Each layer updates your CLS token (more refined information integrated throughout layers)
            - Early layers may capture color/light relationships, deeper layers may capture semantic consistencies, etc.
    - **These layers teach the model how different regions of the face relate to each other** (how your brain checks if the eyes match, if the mouth fits the face shape, etc.).
5. Facial Region Attention (**optional** addition)
    - Mechanism to give more focus to areas often manipulated in deepfakes (eyes, mouth, jawline, etc.).
    - Can be done by adjusting attention weights for those patch regions using face landmarks.
    - Helps the model pay more attention to where fakes generally break down.
6. Global Feature Extraction

    - After all attention layers, the CLS token (average of all patches) is used as a **global image summary**.
    - CLS becomes the ViT's **feature vector (512 values)**, representing what it learned about the entire face.
7. Classification Head
    - Global 512-D vector is passed through a small fully connected (dense) layer that outputs:
        - **Real** or **Fake** label
        - Confidence score (0–1)
    - During fusion, this vector will be combined with the CNN and FFT feature vectors.
8. Output Layer
    - Outputs:
        - Feature vector (512 numbers) for fusion with CNN and FFT.
        - Attention heatmap for visualization

***Self-attention weights** computed within the ViT layers can be visualized as **heatmaps** to highlight which regions of the image most influenced the model's prediction - values from here may be joined with other branches to develop one comprehensive heatmap*

FFT - Model Architecture Diagram



FFT - Diagram Explanation

1. Input Layer

   Shape Handling:

   - Same as the CNN and ViT branches - accepts the preprocessed 224×224×3 image (already face-aligned, cropped, and normalized).

2. RGB to Grayscale Conversion (optional)

   To make computation easier, emphasize frequency intensity over color, the image can be converted from RGB to grayscale.

   Color often adds redundancy in frequency analysis.

   - Grayscale retains the essential intensity patterns for detecting frequency-based artifacts.

   Output shape: **(224 × 224 × 1)**

3. Apply 2D Fast Fourier Transform (FFT)

   Applies a 2D FFT to the image to convert it from the spatial domain (pixel intensities) into the frequency domain.

   This produces two main components:

   - **Magnitude spectrum**: shows how strong each frequency component is.
   - **Phase spectrum**: shows spatial alignment, often less useful for artifact detection.

   Deepfakes often show unnaturally high or low-frequency distributions, which this step makes visible.

   - Output: Frequency map **(224 × 224 × 1)**

4. Frequency Shift (FFT Shift)

   Centers the zero-frequency (lowest) component of the FFT image using **fftshift()**.

   Makes visualization and feature extraction easier by placing low frequencies in the center and high frequencies near edges.

5. Log-Scale Normalization

   Applies log scaling to compress large intensity differences and enhance subtle patterns:

   - **log (1 + |FFT|)**

   Helps reveal faint periodic artifacts that would otherwise be hidden by strong frequency components.

   Output: Log-scaled FFT image **(224 × 224 × 1)**

6. Frequency Feature Extraction (via CNN)

   A lightweight CNN extracts local frequency-domain patterns from the FFT magnitude map.

   Possible layers:

   - Conv2D(64) → ReLU
   - MaxPooling(2×2)
   - Conv2D(128) → ReLU
   - Conv2D(256) → ReLU
   - MaxPooling(2×2)

These layers detect:

- Missing or weak high-frequency details (blurriness)
- Overly smooth regions (low variance)
- Repetitive or checkerboard patterns (GAN noise)

Output**:** Frequency feature map **(56 × 56 × 256)**

7. Global Average Pooling
   Compresses feature map into a single global feature vector by averaging over all frequency regions.
   This **retains the most dominant frequency features** while reducing dimensionality.
   Output: **(1 × 1 × 512)**
8. Fully Connected Layer (Dense)
   Transforms the global frequency feature vector into a fixed-length representation
   - compatible with CNN and ViT outputs.

   Used to refine/scale features before fusion.
9. Output Layer
   512-D Feature Vector (z_fft): sent to the fusion module with CNN + ViT.
   Optional Frequency Heatmap - can be computed from the **final CNN feature maps** using Grad-CAM to highlight frequency regions most influential in detecting deepfake artifacts.

Final Fusion Model (combines the three distinct model outputs into a single classification)